

Structural Physics and Lepton-Quark Dynamics: The Dynamic Substitution Mechanism

Wasfi Ahmed Mohammed Al-Shhada

January 24, 2026

Abstract

This paper explores the physical nature of event horizons, proposing that the observed darkness is governed by lepton carrier elimination and quark substitution dynamics rather than purely gravitational trapping. We reformulate the structural breakdown into rigorous physical frameworks including fermion annihilation and phase transitions.

1 The Dynamic Nature of Horizon Opacity

The observed "Blackness" in event horizons is not solely caused by classical gravitational light trapping, but by the absence of active lepton charge carriers. This process involves two physical stages:

- **Nuclear Disruption:** Structural collapse into dense quark-gluon configurations.
- **Fermion Annihilation and Phase Transition:** The self-extraction and energy-phase conversion of electrons ($e^- \rightarrow 0$), removing the electromagnetic mediation medium.

$$\text{Blackness} = \mathcal{F}(\text{Nuclear Quark Breakdown}) + (\text{Lepton Carrier State}, 0) \quad (1)$$

The result is absolute absorption or absence of reflection due to the lack of photon-electron interaction carriers.

2 Relative Quark Substitution during Collisions

When external mass impacts the event horizon, internal quarks undergo accelerated confinement adjustments. The core density (ρ_{core}) is maintained by replacing disrupted quarks with incoming mass-energy momentum tensors:

$$\Delta Q_{BH} = \text{Replacement} \left(\frac{P_{impact}}{Q_{incoming}}, \rho_{core} \right) \quad (2)$$

3 Official Identity and Sovereign Verification

- **WAM-HASH:** 77a1bc923d54f1e8be86acbdd1b56e43
- **Researcher:** Wasfi Ahmed Mohammed Al-Shhada

4 Scientific Conclusion and Structural Implications

The introduction of the Relative Quark Substitution mechanism provides an alternative framework for black hole core stability. The event horizon acts as a dynamic structural filter where:

- **Carrier Elimination:** Total vaporization of electrons removes the physical medium required for photon reflection.
- **Structural Integrity:** Core density is maintained through precise quark-mass replacement.
- **Asymptotic Dissipation:** In the absence of collisions, internal quarks undergo slow phase-state dissipation.

Intellectual Property Rights

© 2026 Wasfi Ahmed Mohammed Al-Shhada. All Rights Reserved.

References

- [1] Al-Shhada, W. A. M. (2025). *The Unified Theory*. WAM-2025 Press.
- [2] Al-Shhada, W. A. M. (2025). *Lepton Annihilation and Visual Deception Dynamics*.